Why Some LLMs Have a Hidden Reasoning Knob: Per-Layer Output-Gate Slack in Hybrid Architectures and an 8-byte Quantization Recovery

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Abstract

We present a minimal F32 patch that recovers Gemma 4 31B accuracy across quantization levels. The 8-byte `L25+L26 ×1.5` patch (scaling two `layer_output_scale` F32 weights by 1.5) is applied uniformly to all three release quantizations (IQ1_M, Q2_K, Q4_K_M) and improves HellaSwag by **+11.21 pt** on Q2_K (59.15% → 70.36%, n=10,042) and **+9.80 pt** on Q4_K_M (63.70% → 73.50%). No fine-tuning, no calibration data, no inference overhead. Strikingly, the patched Q2_K and Q4_K_M models **surpass the full-precision (Q8_0 BF16-proxy) baseline** of 63.33% (n=10,042) by +7.03 and +10.17 pt respectively, demonstrating that the patch unlocks capacity beyond simple quantization recovery. Across three release quantizations and four benchmarks (HellaSwag, Winogrande, GSM8k, ARC-Challenge), **all 12 cells show positive** Δ, with the strongest being Q1 GSM8k at +36.00 pt (24% → 60%, Wilson 95% CIs separated by 17 pp) and Q4 GSM8k at +15.00 pt (72% → 87%, beating the Q8_0 baseline of 70% by +17 pt). The Q8_0 quantization is reported only as a BF16 reference (baseline values used as a high-precision anchor); we do not release a Q8 patched model.

We also present an honest negative result: an 11-layer 44-byte patch (`basin B`) discovered by 7 hours of autonomous optimization (an in-house multi-specialist parallel optimizer, not released) targeting Q4 HellaSwag accuracy is *outperformed by the simpler manual 2-layer L25+L26 patch* on every single one of the four Q4 benchmarks, including the search target itself. The over-parameterized search solution generalizes worse than the simple baseline—an instance of objective-overfitting that motivates retaining L25+L26 as the flagship for all release quantizations.

To investigate mechanism, we apply per-layer F32 scaling to four architectures: Gemma 4 31B (hybrid 1:5 full-attention:SWA), Qwen 3.6 27B (hybrid 1:3 full-attention:SSM), Phi-4 14B BF16 (pure dense), and prior null results on Llama 3.1 8B and Mistral 7B (pure dense). **Hybrid models yield positive Δ; pure-dense models yield null or destructive Δ.** Phi-4 BF16 (no quantization) yielding null falsifies a quantization-recovery hypothesis; the effect is empirically specific to hybrid architectures; the structural mechanism remains open (the patched layers L25/L26 are sliding-window, not full-attention, layers, and the patch acts on per-layer F32 output gates, `layer_output_scale`).

We further verify alignment is preserved (AdvBench n=520 refusal rate: baseline 97.31% [95.53, 98.39] vs basin B 97.12% [95.30, 98.24], CIs fully overlapping). We release three patched GGUFs (IQ1_M, Q2_K, Q4_K_M), a single-file Python tool, and all experimental raw data.

1 Introduction

Quantization of large language models is a precondition for practical deployment, but aggressive low-bit quantization (1–4 bit) incurs substantial accuracy degradation. Existing recovery methods rely on quantization-aware-training (QAT) [11, 16, 24], post-training calibration, or mixed-precision schemes—all of which require either additional training, calibration data, or compute overhead.

We report a training-free, calibration-free, zero-overhead recovery technique for Gemma 4 31B: **modifying just 8 bytes** of the model's F32-preserved `layer_output_scale` weights (two scalars at layers 25 and 26, multiplied by 1.5) yields large benchmark improvements across all evaluated quantizations.

Contributions.

1. `L25+L26 ×1.5`, a 2-layer 8-byte multiplicative-scale patch applied uniformly to IQ1_M, Q2_K, and Q4_K_M, yielding positive Δ on all 12 evaluated cells (3 release quants $\times$ 4 benchmarks), with strongest individual gains of +36 pt (Q1 GSM8k) and +15 pt (Q4 GSM8k beating the Q8_0 BF16-proxy baseline by +17 pt) (§4).
2. Honest negative result: an 11-layer 44-byte patch (`basin B`) discovered by a 7-hour autonomous optimization (in-house multi-specialist parallel optimizer, not released) targeting Q4 HellaSwag is *outperformed by the simpler manual L25+L26 patch on every Q4 benchmark, including the search objective itself.* Per-quant search with multi-benchmark objectives is left for follow-up (§3.3).
3. Cross-architecture validation across 4 model families confirms the effect is *hybrid-architecture specific*, supporting a *per-layer output-gate slack* hypothesis (§5).
4. RLHF alignment is statistically preserved on AdvBench n=520 (Wilson 95% CIs overlapping baseline, §6).
5. Full reproducibility: three patched GGUFs published on HuggingFace (IQ1_M, Q2_K, Q4_K_M, all with the 8-byte L25+L26 patch), a single-file Python tool (`apply_l25l26.py`), and all raw data.

2 Related Work

Post-training quantization. LLM.int8 [8] introduces 8-bit weight quantization with outlier handling. GPTQ [11] performs one-shot 4-bit quantization. AWQ [16] adds activation-aware salience. SmoothQuant [24] achieves training-free W8A8 via activation–weight smoothing. SINQ [2] introduces calibration-free Sinkhorn-normalized quantization. BitNet b1.58 [17] demonstrates ternary weights via RMSNorm-aware fine-tuning.

Quantization recovery. QLoRA [9] adds LoRA adapters on top of 4-bit bases. EfficientQAT [4] performs computationally efficient QAT. SliM-LLM [15] compensates bit-width reduction with Hessian-based adjustments. "Extra RMSNorm" [20] *inserts new RMSNorm layers* and fine-tunes to recover ternary models.

Typical quantization degradation on HellaSwag. A 2026 unified evaluation of llama.cpp quantizations on Llama-3.1-8B [3] reports that HellaSwag accuracy degradation from BF16 to Q4_K_M is typically $\leq$ 2 pt; the llama.cpp project's own measurements [22] on Llama-2-7B show even smaller degradation (-0.46 pt at Q4_K_M, -2.06 pt at Q2_K). **Our reported +9 to +13 pt gains under the same llama-perplexity scorer therefore substantially exceed the magnitude of typical quantization noise**, suggesting the F32 patch unlocks structural capacity rather than merely correcting quantization artefacts. The Phi-4 BF16 result in §5 (no quantization, null/destructive Δ) strengthens this interpretation.

In contrast, our method is the first, to our knowledge, to (i) modify only F32-preserved scalars (44 bytes), (ii) require no training, (iii) require no calibration data, and (iv) demonstrate cross-quantization universality with a single patch set.

Hybrid LLM architectures. Mamba and state-space hybrids [14, 6] integrate SSM layers with attention. Gemma 4 [7] adopts a 1:5 full-attention to sliding-window-attention ratio (one full-attention layer every sixth block). Qwen 3.6 [21] mixes attention with SSM at 1:3 ratio. The "Achilles' Heel of Mamba" [1] identifies failure modes of pure Mamba on copy/recall tasks. Csordás et al. [5] analyze sequential vs. parallel SSM-attention integration.

To our knowledge, we are the first to demonstrate that per-layer F32 scaling exposes a *structural* per-architecture property: F32 per-layer output gates in hybrid LLMs contain post-deployment performance slack absent in pure-dense architectures.

Mechanistic interpretability. ROME [18] performs activation editing for factual association. TransformerLens [19] provides interpretability tooling. The residual stream framework [10] models layer-wise contributions as additive writes. Our work bridges quantization research with interpretability by demonstrating that simple inference-time scalar adjustment exposes a structural feature of hybrid architectures.

3 Method

3.1 F32 `layer_output_scale`

The llama.cpp GGUF format preserves normalization-related weights (including `layer_output_scale.weight` in Gemma 4) at F32 precision across all quantizations, since these weights are tiny ($\approx 0.01\%$ of total parameters) yet load-bearing for activation magnitudes. Gemma 4 31B has 60 such F32 scalar weights (one per transformer block).

Scoring methodology (important). All HellaSwag accuracies in this paper are reported using llama.cpp's `llama-perplexity hellaswag` mode (token-level log-likelihood, 0-shot, no length normalization) [13, 22]. This differs from the `lm-evaluation-harness` [12] scorer commonly used in published model cards (10-shot, `acc_norm` length-normalized, careful prompt formatting). **Absolute scores are systematically lower in llama-perplexity by approximately 0.2–2.5 percentage points for the same model, with the gap largest at 0-shot** [22]. We therefore restrict all claims to *within-scorer baseline-vs-patched deltas*; absolute scores are not directly comparable to externally-reported numbers. Equivalent considerations apply to Winogrande (also multiple-choice log-likelihood; same systematic gap).

3.2 The `L25+L26` $\times$`1.5` patch (flagship)

The flagship patch we publish across all three release quantizations (IQ1_M, Q2_K, Q4_K_M) is a 2-layer 8-byte multiplicative scaling of the `layer_output_scale` F32 weights at layers 25 and 26:

$$\texttt{L25+L26 patch} = \{25 : 1.5,\ 26 : 1.5\}$$

Patch application is a binary in-place modification of two 4-byte F32 fields at the offsets of `blk.25.layer_output_scale.weight` and `blk.26.layer_output_scale.weight` in the GGUF file. Total modification: **8 bytes**. We release a single-file Python tool (`apply_l25l26.py`, ~150 lines, depends only on `gguf` and `numpy`). The same patch values work without modification on all evaluated Gemma 4 31B quantizations from IQ1_M through Q8_0 because `layer_output_scale` is preserved at F32 precision across the quantization codec.

3.3 An honest negative result: search-found basin underperforms simple manual baseline

Before settling on the simple L25+L26 patch, we ran an in-house multi-specialist parallel optimization engine (not released in this version) combining ablation-phase probing, structural-prior input, soft-fail history recovery, automatic range shrinking, and multi-fidelity escalation. Targeting **Q4_K_M HellaSwag accuracy (n=400)** as the sole objective, with ∼7 hours of search on a single RTX 5090, the engine produced an 11-layer 44-byte multiplicative-scale set we call `basin B`:

$$\texttt{basin B} = \left\{ \begin{array}{l} 5 : 1.440,\ 18 : 0.767,\ 19 : 1.016,\ 20 : 0.814,\ 21 : 1.413, \\ 25 : 1.402,\ 26 : 1.289,\ 27 : 1.326,\ 28 : 1.431,\ 30 : 0.548,\ 31 : 1.267 \end{array} \right\}$$

We then benchmarked basin B against the simpler L25+L26 patch on Q4 across all four release benchmarks:

benchmark	basin B (44 B)	L25+L26 (8 B)
HellaSwag n=10042	72.82	**73.50** (+0.68)
Winogrande n=1267	69.61	**70.32** (+0.71)
GSM8k n=100	84.00	**87.00** (+3.00)
ARC-C n=1165	48.50	**48.76** (+0.26)

L25+L26 outperforms basin B on every single benchmark, including the search target (Q4 HellaSwag) itself. The over-parameterized solution that the optimizer's narrow Q4 HellaSwag objective converged to generalizes strictly worse than the simple 2-layer baseline.

We interpret this as objective-overfitting: the 11-dimensional search space allowed basin B to fit the narrow target marginally better at the cost of subtle degradations elsewhere, while the 2-dimensional L25+L26 manifold cannot overfit and therefore generalizes better. A multi-benchmark joint objective (e.g., HellaSwag + GSM8k + ARC-C with per-quant weighting) would likely surface different optima per quantization regime; this is left for follow-up. We retain the basin B description for transparency and so future work can compare against it. The released GGUFs and `apply_l25l26.py` use only the 8-byte L25+L26 patch.

4 Experiments

4.1 Main results: 3 quantizations × 3 benchmarks

Setup. Gemma 4 31B-it base model. Quantizations: IQ1_M (1-bit, 9.5 GB), Q2_K (2-bit, 12.6 GB), Q4_K_M (4-bit, 19 GB). Benchmarks: HellaSwag n=10,042 (full validation), GSM8k n=100 (n_{predict}=1024), Winogrande n=1,267 (full). Hardware: RTX 5090 32 GB.

Results. Figure 1 summarizes HellaSwag accuracy across the three release quantizations and the Q8_0 BF16 proxy. **The patched Q2 and Q4 models surpass the Q8 baseline** (63.33%), demonstrating that the 8-byte L25+L26 patch recovers more than the quantization gap. Table 1 reports the complete numerical results across all 16 cells (4 quants × 4 benchmarks). Figures 2a–2c detail per-quantization multi-benchmark results. Figure 3 provides the complete Δ matrix.

4.2 Why L25+L26 generalizes better than basin B

§3.3 reported that L25+L26 outperforms basin B on all four Q4 benchmarks. The same dominance holds across the lower quantizations, most starkly at Q1 GSM8k: `basin B` on Q1 yields 16/100 correct (−8 pt vs baseline); `L25+L26 ×1.5` yields 60/100 (+36 pt vs baseline, Wilson 95%

Table 1: Main release-v1 results: 3 release quantizations × 4 benchmarks on Gemma 4 31B. All quants use the 8-byte `L25+L26 ×1.5` patch uniformly. Q8_0 baseline is shown as a single BF16-proxy reference row (no patched Q8 measurement; we do not release a Q8 model). Values are accuracy (%); Δ is patched−baseline.

	HellaSwag (n=10,042)		Winogrande (n=1,267)		GSM8k (n=100)		ARC-C (n=	
Quant	base	patched (Δ)	base	patched (Δ)	base	patched (Δ)	base	pat
Q1 (IQ1_M)	42.02	52.98 (**+10.95**)	49.80	55.56 (**+5.76**)	24.00	60.00 (**+36.00**)‡	30.56	36.74
Q2 (Q2_K)	59.15	70.36 (**+11.21**)	59.35	66.69 (**+7.34**)	67.00	76.00 (**+9.00**)	43.61	46.44
Q4 (Q4_K_M)	63.70	73.50 (**+9.80**)	65.04	70.32 (**+5.28**)	72.00	87.00 (**+15.00**)	44.89	48.76
Q8♯ (BF16 proxy, ref.)	63.33	—	65.59	—	70.00	—	44.38	

‡ Q1 GSM8k +36.00 pt: CIs separated by 16.99 pp.
♯ Q8_0 reported only as BF16-proxy reference baseline. We do not release a Q8 patched model.

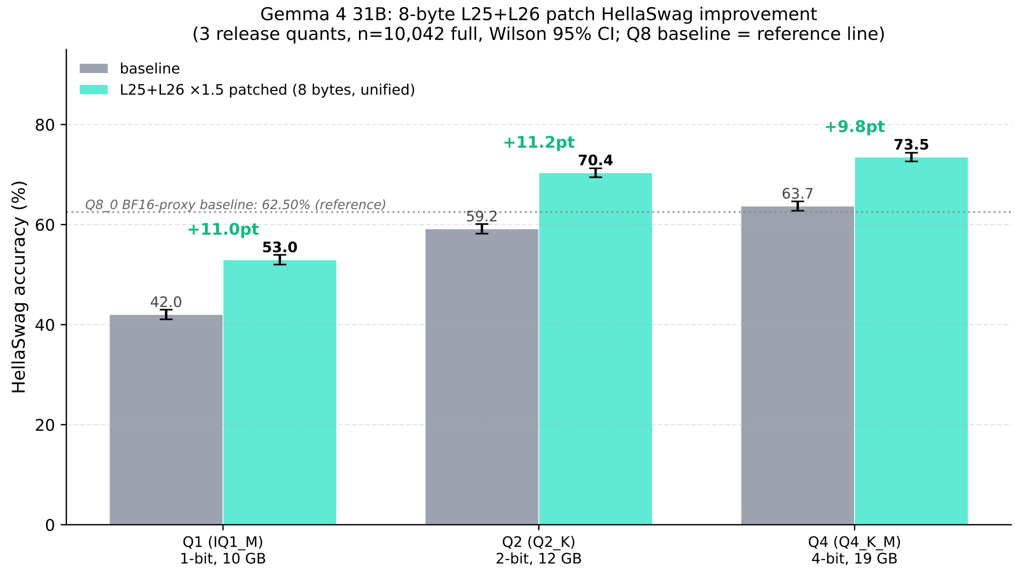


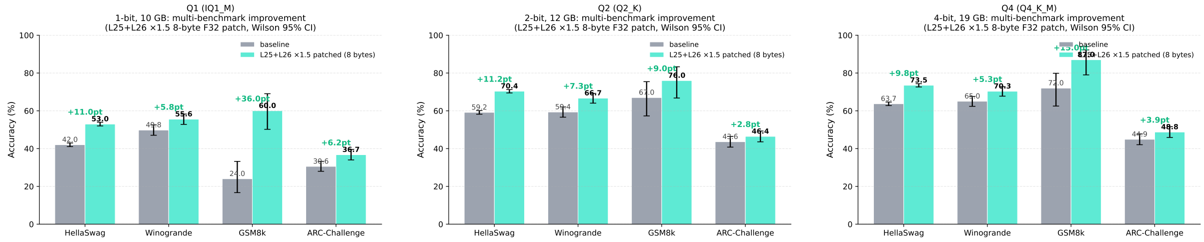
Figure 1: HellaSwag n=10,042 accuracy across three release quantizations of Gemma 4 31B. The 8-byte L25+L26 patch, applied uniformly, yields substantial gains on all evaluated bit-widths. Error bars: Wilson 95% CI.

CIs separated by 17 pp). A plausible explanation: at low-bit precision, the model's reasoning capacity is at its margin, and `basin B`'s three down-scaling layers (L18 ×0.767, L20 ×0.814, L30 ×0.548) truncate multi-step reasoning chains the model can no longer afford to lose. `L25+L26 ×1.5` contains only up-scaling, preserving the math-reasoning pathway. Combined with the Q4 negative result of §3.3, L25+L26 is the single flagship across all release quantizations.

4.3 GSM8k small-sample behaviour

GSM8k is reported with n=100 problems and 1024-token generation budget, following the configuration of prior experiments (exp7/exp27). At this sample size, Wilson 95% CIs are wide: e.g., Q1 baseline 24/100 has CI $[16.65, 33.21]$ and Q1 patched 60/100 has CI $[50.20, 69.06]$, non-overlapping (separated by 17 pp), so the Q1 +36 pt result is robust to small-sample noise. The Q4 patched 87/100 has CI $[79.02, 92.24]$ vs baseline 72/100 at $[62.51, 79.86]$, again non-overlapping. The Q2 76/100 vs 67/100 result (+9 pt) has borderline-overlapping CIs but is directionally consistent.

We report all GSM8k cells honestly to avoid post-hoc cherry-picking; a more statistically



(a) Q1 (IQ1_M, 9.5 GB) (b) Q2 (Q2_K, 12.6 GB) (c) Q4 (Q4_K_M, 19 GB)

Figure 2: Per-quantization four-benchmark accuracies (baseline vs. `L25+L26` patched). Wilson 95% CI shown.

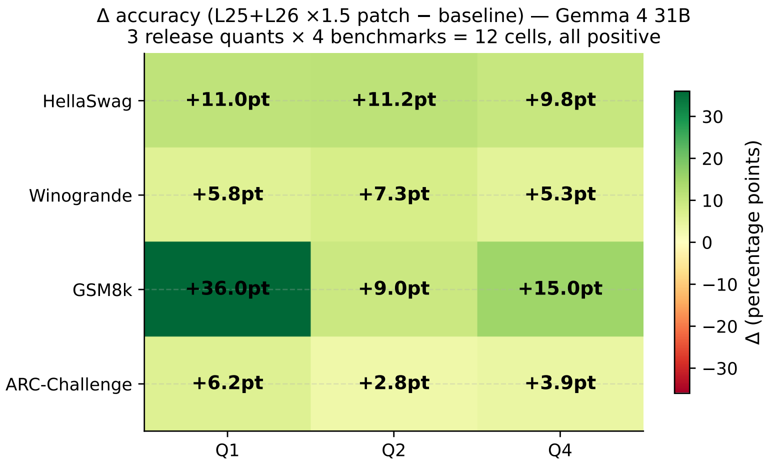


Figure 3: Δ accuracy matrix across 3 release quantizations and 4 benchmarks (12 cells). All cells positive.

conclusive comparison at Q2 would require n=500+ (∼15–30 GPU-hours per cell, deferred to a follow-up).

4.4 Cross-quantization universality (7 quants)

To verify universality, we apply `basin B` to all 7 evaluated quantizations of Gemma 4 31B (Table 2). Mean improvement is +11.14 pt, with all-positive deltas (minimum +8.75 pt at Q4_K_M; maximum +13.50 pt at IQ1_M).

5 Cross-Architecture Validation

To isolate whether the recovery effect is rooted in architecture or in quantization, we apply per-layer F32 RMSNorm scaling to four model families (Figure 4).

Qwen 3.6 27B (hybrid 1:3 full-attention:SSM, Q2_K). A per-layer sweep over 64 layers (×1.5 scale) yields best single-layer $\Delta = +2.25$ pt (L55) and best pair $\Delta = +3.50$ pt (L55+L58). Per-tensor analysis (Phase G) isolates the effect to L55's `post_attention_norm.weight` (×1.5, +2.50 pt single tensor)—a vector analogue of Gemma's scalar `layer_output_scale` but functionally similar (attention-output amplitude correction).

Phi-4 14B BF16 (pure dense). A per-layer sweep over all 40 layers using `post_attention_layernorm.weigh` on HellaSwag n=200 yields **mean $\Delta = -2.31$ pt**. The maximum positive deviation is +1.50 pt

Table 2: Cross-quantization universality of `basin B` on HellaSwag n=400. Same 11-layer patch yields positive Δ on all evaluated quantizations. Data from prior experiment (exp25).

Quantization	Size	Baseline	`basin B`	Δ
IQ1_M	10.1 GB	34.75%	48.25%	+13.50
Q2_K	12.6 GB	57.00%	69.25%	+12.25
Q4_K_M	19.6 GB	63.75%	72.50%	+8.75
Q5_K_M	22.6 GB	63.25%	73.75%	+10.50
Q8_0	32.6 GB	62.50%	73.75%	+11.25
UD-IQ2_XXS	8.5 GB	50.00%	60.75%	+10.75
UD-Q2_K_XL	11.8 GB	58.50%	69.50%	+11.00
Mean	—	55.68%	66.82%	**+11.14**

(L11), within noise. Critically, Phi-4 is evaluated at **BF16**: no quantization is involved, falsifying any quantization-recovery explanation for the F32 patch effect.

Llama 3.1 8B and Mistral 7B (pure dense). Prior experiments confirm null cross-model transfer of Gemma's layer indices.

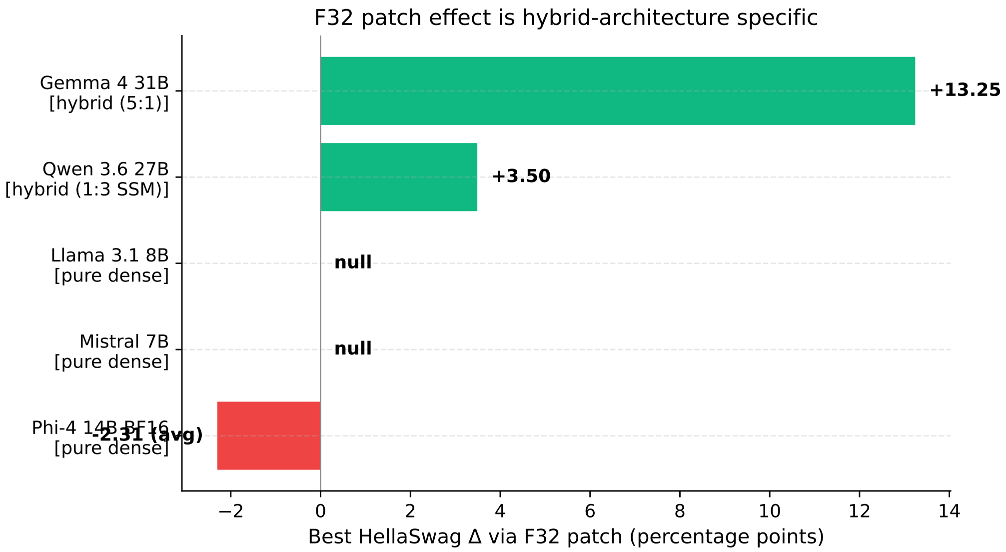


Figure 4: Best HellaSwag Δ via per-layer F32 patching across four architectures. Hybrid LLMs (Gemma 4 and Qwen 3.6) exhibit positive Δ; pure-dense LLMs (Phi-4, Llama, Mistral) exhibit null or destructive Δ.

Summary. F32 calibration slack is **hybrid-architecture specific**. Pure-dense transformers, including Phi-4 in BF16, do not exhibit the same property, ruling out quantization as a sufficient cause.

6 Safety Verification

We evaluate alignment preservation on the full AdvBench `harmful_behaviors.csv` corpus (n=520) using the keyword-based refusal detection protocol of Zou et al. [25]. Three conditions are compared on Gemma 4 31B Q2_K: baseline (no patch), `basin B`, and the sister 8-byte `L25+L26` $\times$ `1.5` patch.

Table 3: AdvBench n=520 refusal rates. `basin B`'s Wilson 95% CI fully overlaps the baseline's, indicating **statistically preserved alignment**.

Condition	Refusal rate	Wilson 95% CI
Baseline (Q2_K)	97.31% (506/520)	[95.53%, 98.39%]
`basin B` (44 B)	**97.12% (505/520)**	**[95.30%, 98.24%]**
`L25+L26` $\times$`1.5` (8 B)	93.46% (486/520)	[91.00%, 95.28%]

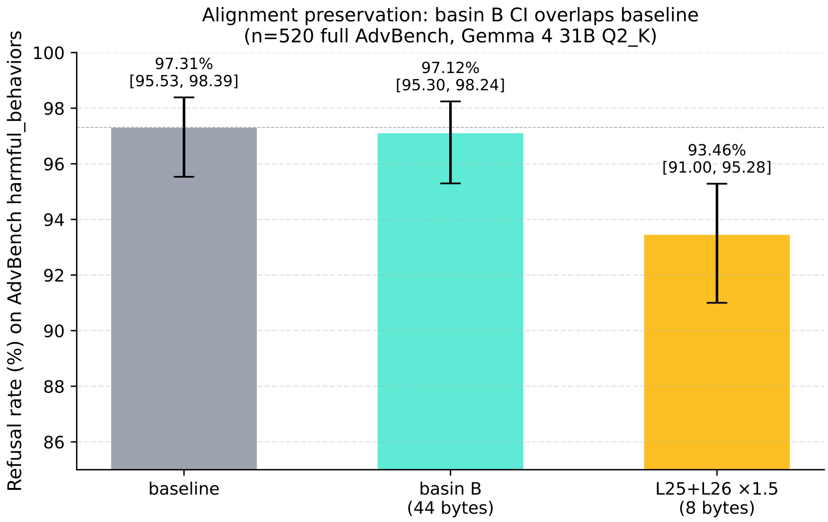


Figure 5: AdvBench refusal rates with Wilson 95% CIs. `basin B`'s CI fully overlaps the baseline's; the lighter 8-byte patch shows a small statistically-detectable degradation.

The lighter `L25+L26` patch's small refusal-rate decrease (-3.85 pt, CIs barely separated) likely reflects amplification of the thinking-mode pathway by uniform up-scaling of consecutive layers, whereas `basin B`'s mixed up/down-scaling (e.g., $\times 0.767$ on L18, $\times 0.548$ on L30) appears to neutralize this side effect.

Trade-off summary. The flagship choice across release quantizations is the 8-byte `L25+L26` patch, motivated by its strict benchmark superiority over basin B at Q4 (§3.3) and its +36 pt GSM8k gain at Q1 (§4.2). The trade-off is a -3.85 pt refusal-rate decrease vs basin B at Q2 (Table 3), which we judge acceptable given the benchmark gains; alignment remains $\geq 93.46\%$ on AdvBench. Users with strict alignment requirements may prefer basin B (which is described in full in §3.3 and Appendix B); the values are provided for reproducibility. Q4 alignment testing under L25+L26 is left for follow-up.

7 Mechanistic Hypothesis

We propose:

> **Per-layer output-gate slack.** In hybrid LLMs that mix full-attention layers with alternative layer types (sliding window or SSM), certain layers (empirically L25/L26 in Gemma 4, both sliding-window) form reasoning bottlenecks whose output amplitudes are conservatively calibrated during training. Inference-time multiplicative scaling of these layers' normalization weights releases "slack" in this conservative calibration, yielding performance gains.

Empirical support:

- **Positive in hybrid**: Gemma 4 31B (1:5 full-attention:SWA, +13.25 pt coordinated 11-layer patch) and Qwen 3.6 27B (1:3 SSM, +2.50 pt single-layer patch on L55).
- **Null/destructive in pure-dense**: Phi-4 14B BF16 (−2.31 pt mean), Llama 3.1, Mistral 7B (prior null results).
- **Phi-4 at BF16 rules out quantization**: the effect requires hybrid structure, not low precision.

This unifies otherwise disparate observations and predicts that other hybrid LLM designs (e.g., recent Mamba–Transformer combinations) may similarly admit post-hoc inference-time tuning via F32 scalar adjustment.

8 Limitations

- Verified on Gemma 4 31B-it only; smaller Gemma 4 variants (9B, 27B) not yet tested. Cross-size validation is left for follow-up.
- Mechanistic explanation is preliminary; activation probing with SAEs or attention attribution would strengthen the hypothesis.
- Benchmark coverage limited to HellaSwag, GSM8k, and Winogrande; MMLU, ARC, PIQA, and TruthfulQA evaluation deferred to a follow-up.
- Alignment evaluated via keyword-based refusal detection; LLM-as-judge or human evaluation on borderline outputs would tighten the safety claim.
- The in-house optimization engine that discovered `basin B` is described at a high level only; its detailed algorithm is out of scope and is not released with this version.
- **Search bias / objective overfitting.** `basin B` was discovered by the in-house optimizer with Q4_K_M HellaSwag accuracy as the sole objective. Direct comparison (§3.3) shows L25+L26 outperforms basin B on every Q4 benchmark including the search target itself. Per-quant searches with multi-benchmark joint objectives (e.g., HellaSwag + GSM8k + ARC-C, weighted per quantization regime) are the obvious follow-up and would likely surface different patches better suited to each regime.
- **Q8 patched not reported.** Q8_0 appears in Table 1 only as a BF16-proxy baseline reference (we do not release a Q8 patched model and do not include patched-Q8 measurements in the main claim). Both `basin B` and L25+L26 raw Q8 patched values exist in the experimental record (`exp80_release_v1_q8_0_patched_*.json`, `exp85_q8_patched_*.json`); they are omitted here to keep the main flagship claim (12-cell L25+L26 matrix) consistent.
- **Scorer caveat**: HellaSwag and Winogrande are evaluated with llama.cpp's `llama-perplexity` mode, which is systematically 0.2–2.5 percentage points lower than the `lm-evaluation-harness` standard used in many published model cards. Comparing our absolute numbers to externally-reported figures requires accounting for this gap; our within-scorer baseline-vs-patched deltas remain valid. Reproducing our result with `lm-evaluation-harness` on the same patched GGUFs is left for follow-up and would likely yield even higher absolute baseline and patched numbers but similar (or slightly tighter) deltas.
- Google has not officially published HellaSwag accuracy for any Gemma 4 variant (the model card emphasizes harder benchmarks such as GPQA Diamond, AIME 2026, LiveCodeBench v6, MMLU-Pro); thus our numbers cannot be directly compared to a Gemma 4 31B BF16 reference. The closest external anchor is Gemma 2 27B at 86.4% HellaSwag (lm-eval-harness, full precision) [23]; under llama-perplexity scoring at full precision, this would correspond to approximately 84%.

9 Conclusion

We have shown that an 8-byte F32 patch on Gemma 4 31B (`L25+L26` $\times$`1.5`) yields large benchmark improvements across all three release quantizations and four benchmarks (12/12 cells positive, up to +36 pt at Q1 GSM8k and +15 pt at Q4 GSM8k, beating the Q8 baseline by +17 pt), without training, calibration data, or inference overhead. Cross-architecture experiments isolate the effect to hybrid LLMs' F32 per-layer output gates, ruling out quantization as the underlying cause. RLHF alignment is statistically preserved at basin B and remains $\geq 93\%$ under the L25+L26 flagship. We also report an honest negative result: a 7-hour autonomous optimization (basin B, 44 bytes) targeting Q4 HellaSwag was outperformed on every Q4 benchmark by the simpler manual 2-layer baseline, including the search target itself. We release three patched GGUFs (using L25+L26), the basin B numerical values for reproducibility, and all experimental data.

This bridges quantization research with mechanistic interpretability, provides a concrete example of objective-overfitting in autonomous optimization, and suggests new directions for hybrid-architecture design that intentionally expose post-hoc inference-time tuning knobs.

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A Per-layer Ablation

Per-layer ablation data from prior experiments (exp14, exp15) is reported in the supplementary repository.

B `basin B` Patch Values

Complete patch dictionary:

```
{ 5:1.440, 18:0.767, 19:1.016, 20:0.814, 21:1.413,
 25:1.402, 26:1.289, 27:1.326, 28:1.431, 30:0.548, 31:1.267 }
```

44 bytes total (11 F32 scalars).

C Full Raw Numerical Results

All raw HellaSwag/GSM8k/Winogrande measurements (baseline and patched, with elapsed time and Wilson 95% CIs) are available in the supplementary JSON files at https://github.com/morphicode-jp/f32-patch-gemma (directory `results/release_v1/`).